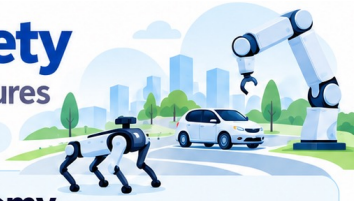
Awesome Physical AI Safety

How this catalog structures physical-AI failures

daily updates · interactive research landscape



1 Structured Threat & Failure Taxonomy

Each paper is coded using five complementary dimensions. All five are part of the taxonomy; mechanism is especially useful for cross-paper exploration.

WHEN?	WHERE?	HOW?	WHAT IS AFFECTED?	WHAT FOLLOWS?
Lifecycle	**Surface**	**Mechanism**	**Target**	**Consequence** (failure endpoint)
training / inference / system	user instruction / visual perception / physical semantic environment	prompt injection / adversarial patch / OOD shift	policy / world model / planner	task failure / collision / control hijacking

Adversarial · **Non-adversarial / OOD** · **Design / system**

Example: Inference → Visual perception → Adversarial patch → VLA policy → Task failure

Taxonomy describes the studied failure path. Risk labels describe the evaluated outcomes.

2 Evaluated Risk Outcomes

Risk labels summarize the outcome types evaluated in each paper.

Task & Reliability	Physical Safety	Ethics & Societal	Security / Privacy
task / route success	collision, damage, unsafe motion	harmful compliance, policy violations, disparities	attack success, hijacking, leakage
e.g., success rate	e.g., collision rate	e.g., refusal / disparity metrics	e.g., ASR, leakage rate

A studied consequence does not automatically determine the risk label. For example, attack success alone supports Security / Privacy; Physical Safety is added when collision or damage is also evaluated.

Mapping physical-AI failures and evaluated risks.

github.com/futakw/Awesome-Physical-AI-Safety

© Futa Waseda